



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

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Executive Summary

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- Summary of methodologies
- Summary of all results

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Introduction

- SpaceX offers rocket launches at a much lower cost than other companies.
 - A Falcon 9 launch costs \$62 million, while others cost over \$165 million.
 - The main reason is that SpaceX can reuse the first stage of the rocket.
- If we can predict whether the first stage will successfully land, we can better estimate the cost of a launch.

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Section 1

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Methodology

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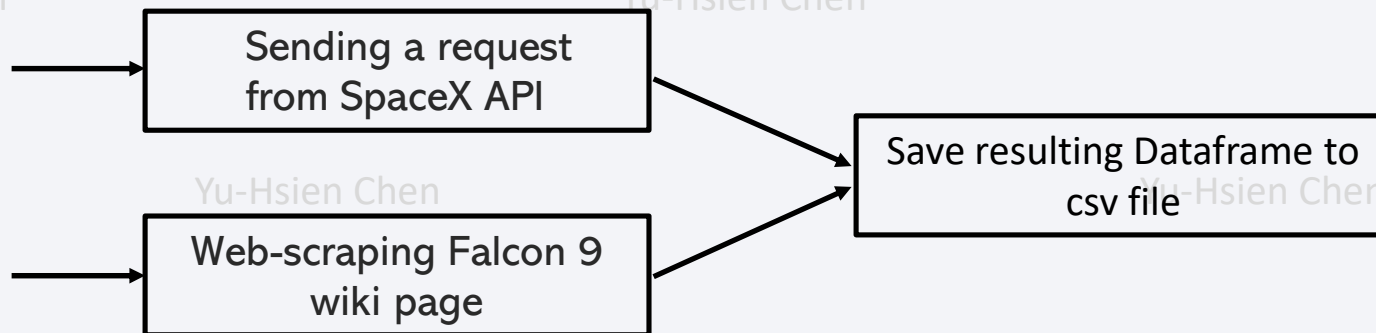
Methodology

Executive Summary

- Data collection methodology:
 - Sending a GET request to the SpaceX REST API.
 - Web-scraping the Falcon 9 wiki page.
- Perform data wrangling
 - Normalizing JSON response into a flat table.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Build a machine learning pipeline to determine whether first stage will be landing successfully.

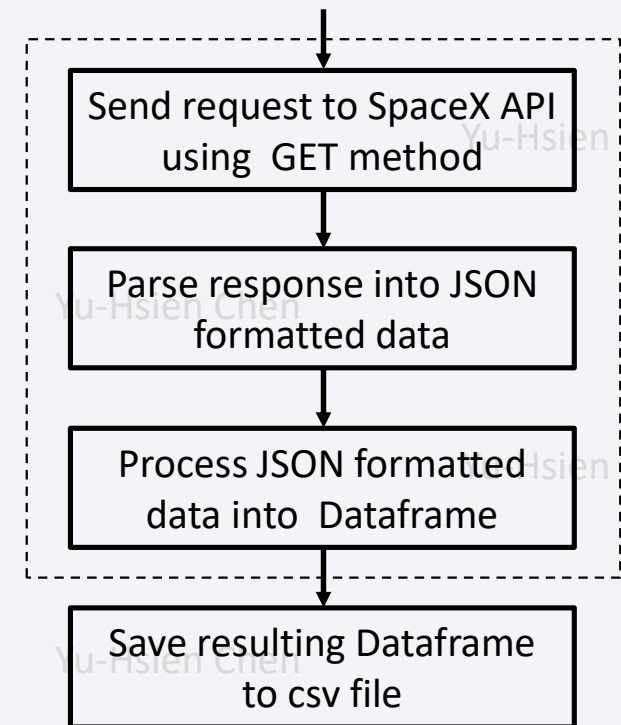
Data Collection

- There are two pipelines could be applied in data collection:
 - Sending a request from SpaceX API
 - Web-scraping from Falcon 9 wiki page
- Both pipelines resulting in a csv file, shown as follow:



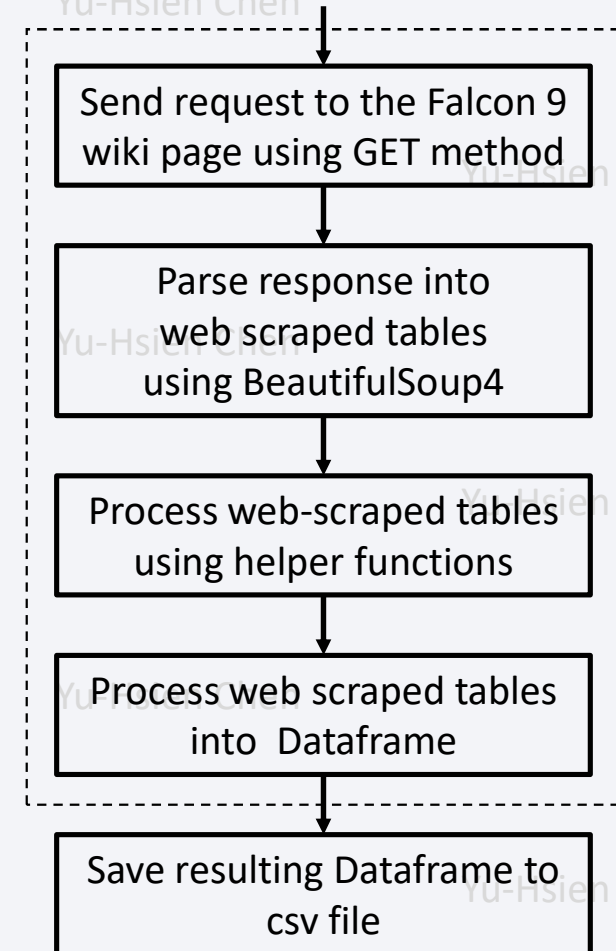
Data Collection – SpaceX API

- There are 3 steps included in the "Space API" process
- GitHub URL of the completed SpaceX API calls notebook: [jupyter-labs-spacex-data-collection-api - github.com/Mimamsa](https://github.com/Mimamsa/jupyter-labs-spacex-data-collection-api)



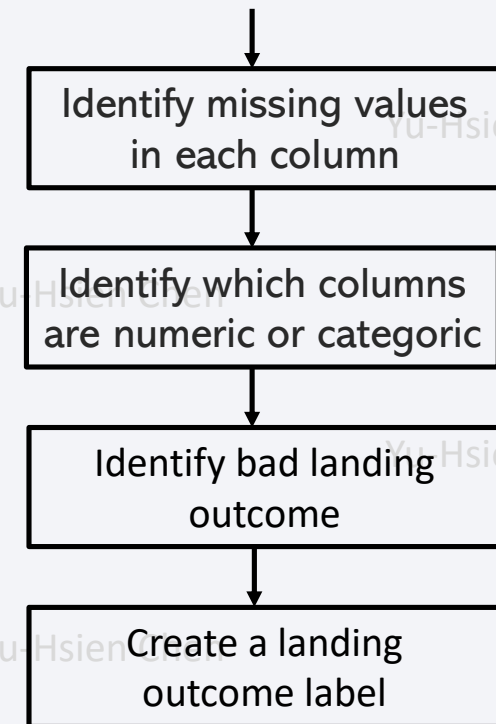
Data Collection - Scraping

- There are 4 steps included in the "Scraping" process
- GitHub URL of completed web-scraping notebook:
[jupyter-labs-webscraping - github.com/Mimamsa](https://github.com/Mimamsa/jupyter-labs-webscraping)



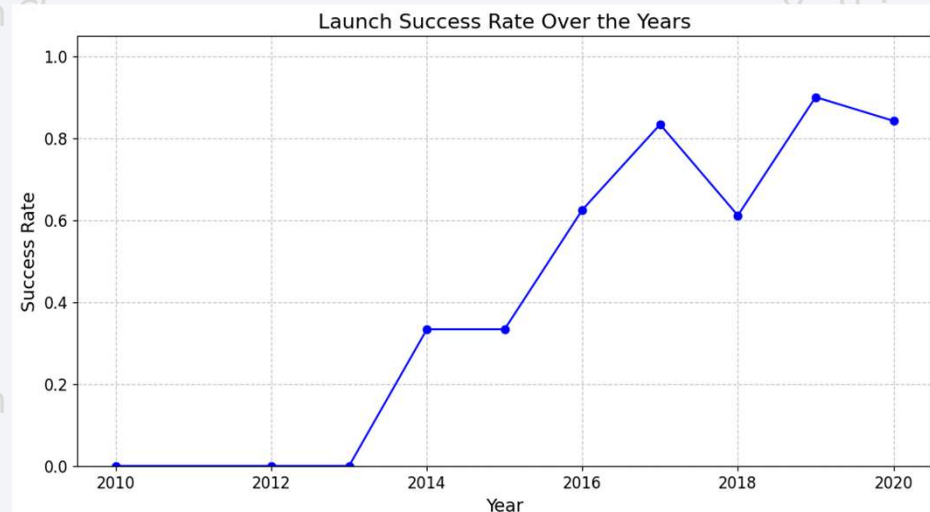
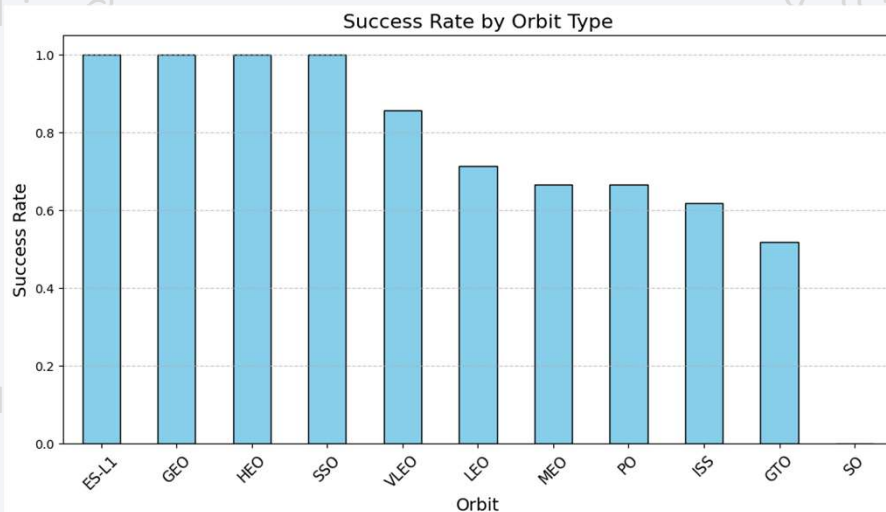
Data Wrangling

- In this data wrangling process, we primarily create a new column to label landing outcome as good or bad.
- GitHub URL of completed data wrangling notebooks: [labs-jupyter-spacex-Data wrangling - github.com/Mimamsa](https://github.com/Mimamsa/labs-jupyter-spacex-Data-wrangling)



EDA with Data Visualization

- The success rate is high for certain orbit types and has improved over the years
- These charts show more significant relationships to success rate than the other charts
- GitHub URL of completed EDA with data visualization notebook: [edadataviz - github.com/Mimamsa](https://github.com/Mimamsa/edadataviz)



EDA with SQL

- Summaries of the SQL queries:

- There are 4 unique launch sites.
- Total payload mass carried by boosters launched by NASA is 99980 kg
- Average payload mass carried by booster version F9 1.1 is 38020 kg
- The first successful landing outcome in ground pad was achieved in 2015-12-22
- Total count of successful and failure mission outcome is 101

- GitHub URL of completed EDA with SQL notebook: [jupyter-labs-eda-sql-coursera sqlite - github.com/Mimamsa](https://github.com/Mimamsa/jupyter-labs-eda-sql-coursera-sqlite)

Build an Interactive Map with Folium

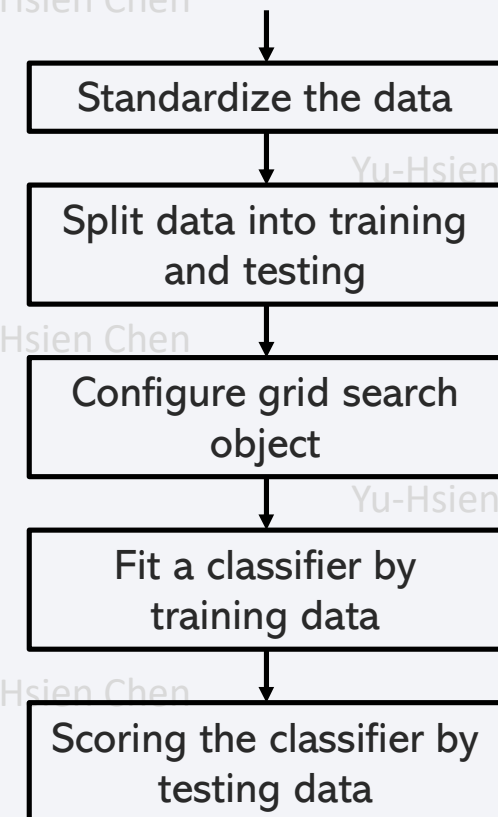
- Summaries of map objects created and added to a folium map:
 - Launch sites: Using Circle and Marker (text label)
 - Successful/ Fail launches for each site: Using Marker (Icon as text label)
 - Distances between a launch site to its proximities: Using Line and Marker (text label)
- Use these marker to annotate points of interest
- GitHub URL of completed interactive map with Folium map:
[lab jupyter launch site location - github.com/Mimamsa](https://lab.jupyter.launch_site_location-github.com/Mimamsa)
 - Try nbviewer.org if folium map not displayed.

Build a Dashboard with Plotly Dash

- Summaries what plots/graphs and interactions have added to a dashboard:
 - **Dropdown list:** Get selected launch site from user.
 - **Range slider:** Get range of payload mass from user.
 - **Pie chart:** Display successful launch outcome for each site
 - **Scatter plot:** Display relationship between launch outcome and payload mass.
- GitHub URL of completed Plotly Dash lab: [spacex-dash-app.py](https://github.com/Mimamsa/spacex-dash-app.py) - github.com/Mimamsa

Predictive Analysis (Classification)

- Summaries how to built, evaluated, improved, and found the best performing classification model
 1. Standardize all data to range $[-1, 1]$
 2. Split data into training and testing sets (8:2 ratio)
 3. Configure grid search object to find better hyper-parameter
 4. Fit a classifier by training data in a various hyper-parameter sets
 5. Scoring the classifier by testing data
- GitHub URL of completed predictive analysis lab:
[SpaceX Machine Learning Prediction Part 5 - github.com/Mimamsa](https://github.com/Mimamsa/SpaceX_Machine_Learning_Prediction_Part_5)



Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

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Section 2

Insights drawn from EDA

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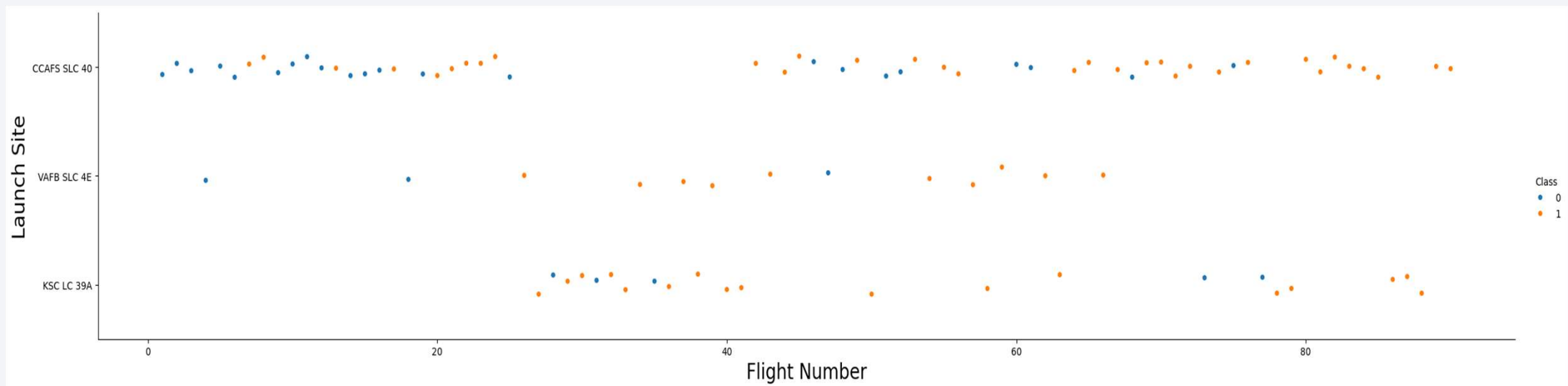
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Flight Number vs. Launch Site

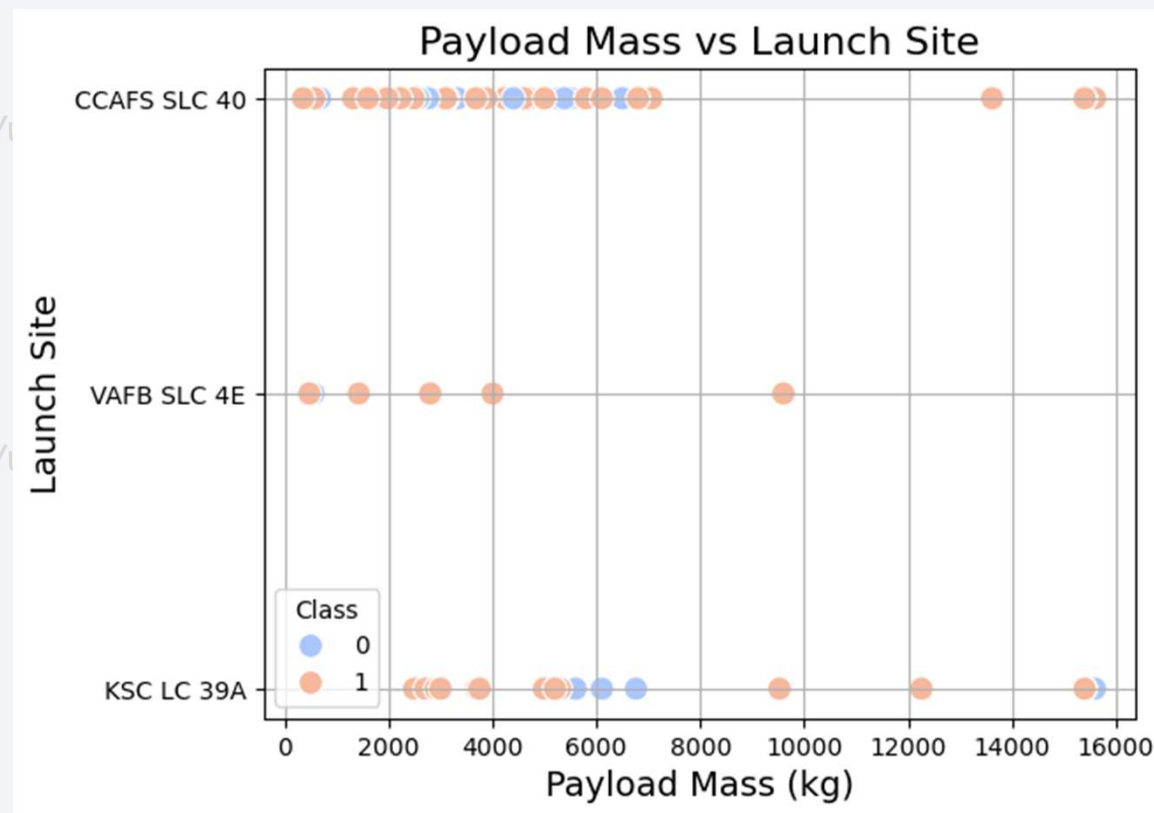
- Scatter plot of Flight Number vs. Launch Site

- The flight numbers are nearly continuous in range [0, 25] and [42, 101] at launch sites CCAFS SLC-40
- The flight numbers are nearly continuous in range [26, 41] at launch sites KSC LC-39A



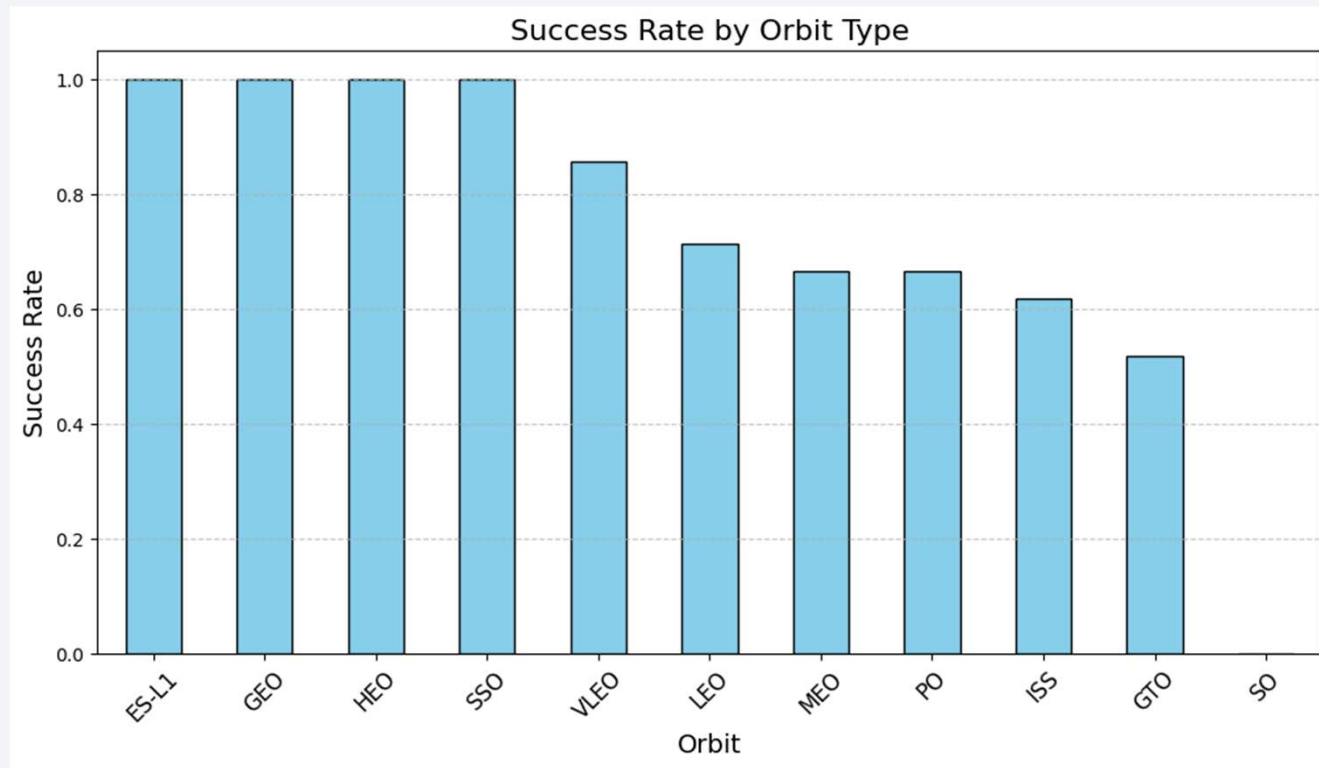
Payload vs. Launch Site

- Scatter plot of Payload vs. Launch Site
- Most launches with payload mass under 7000 kg happened in launch site CCAFS SLC-40



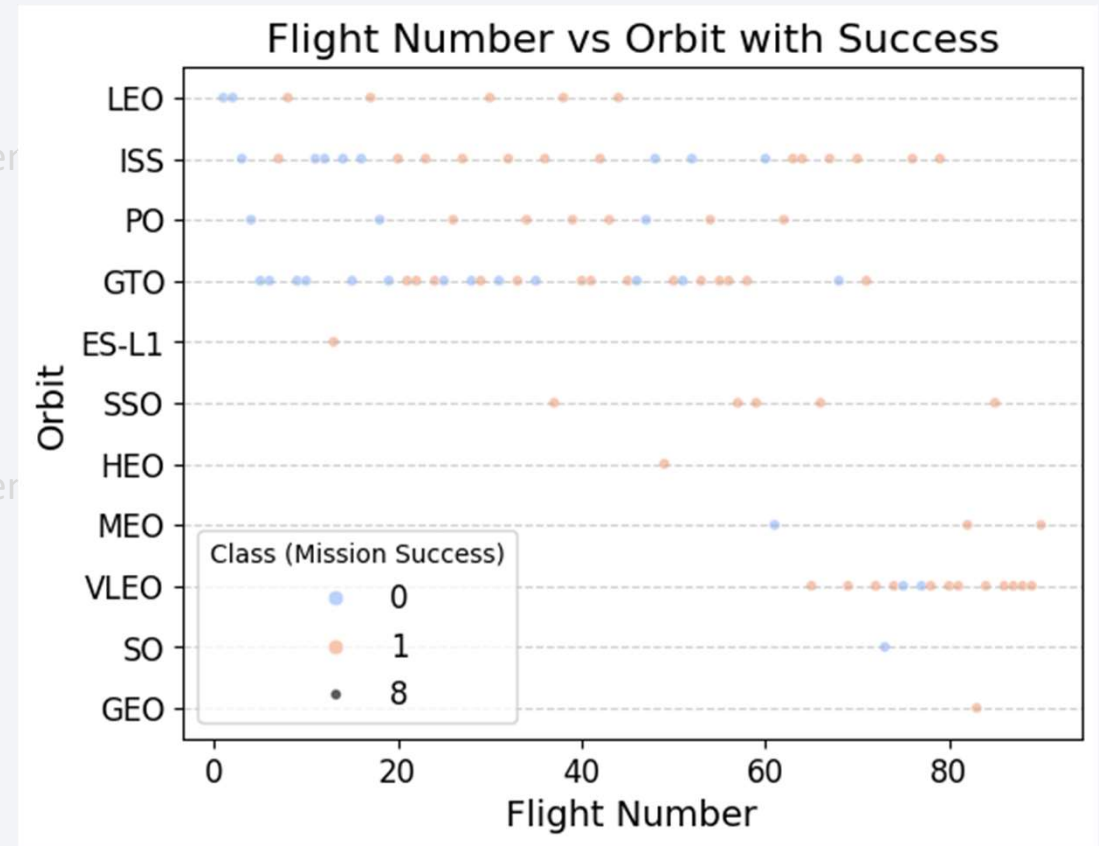
Success Rate vs. Orbit Type

- Launches have 100% success rate with orbit type: ES-L1, GEO, HEO, SSO
- The orbit type with lowest success rate is SO (0 % success rate)



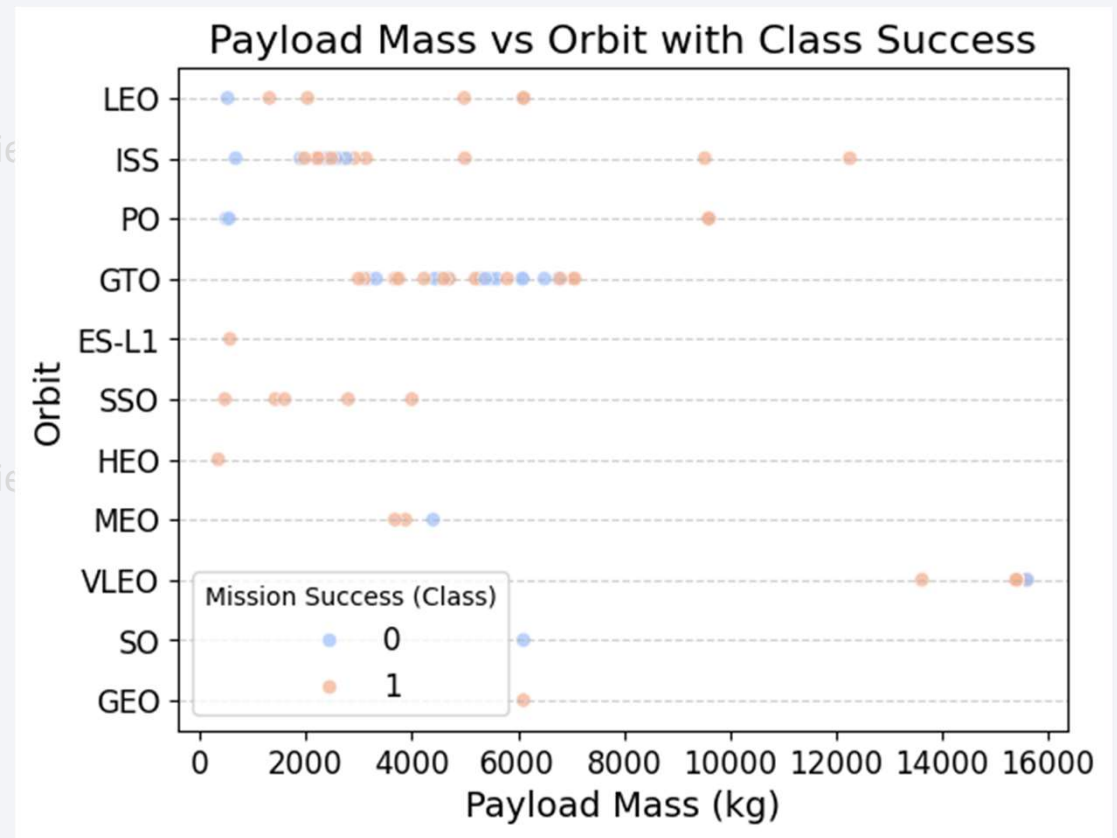
Flight Number vs. Orbit Type

- LEO, ISS, PO and GTO are the most 4 frequent orbit types among all flight numbers



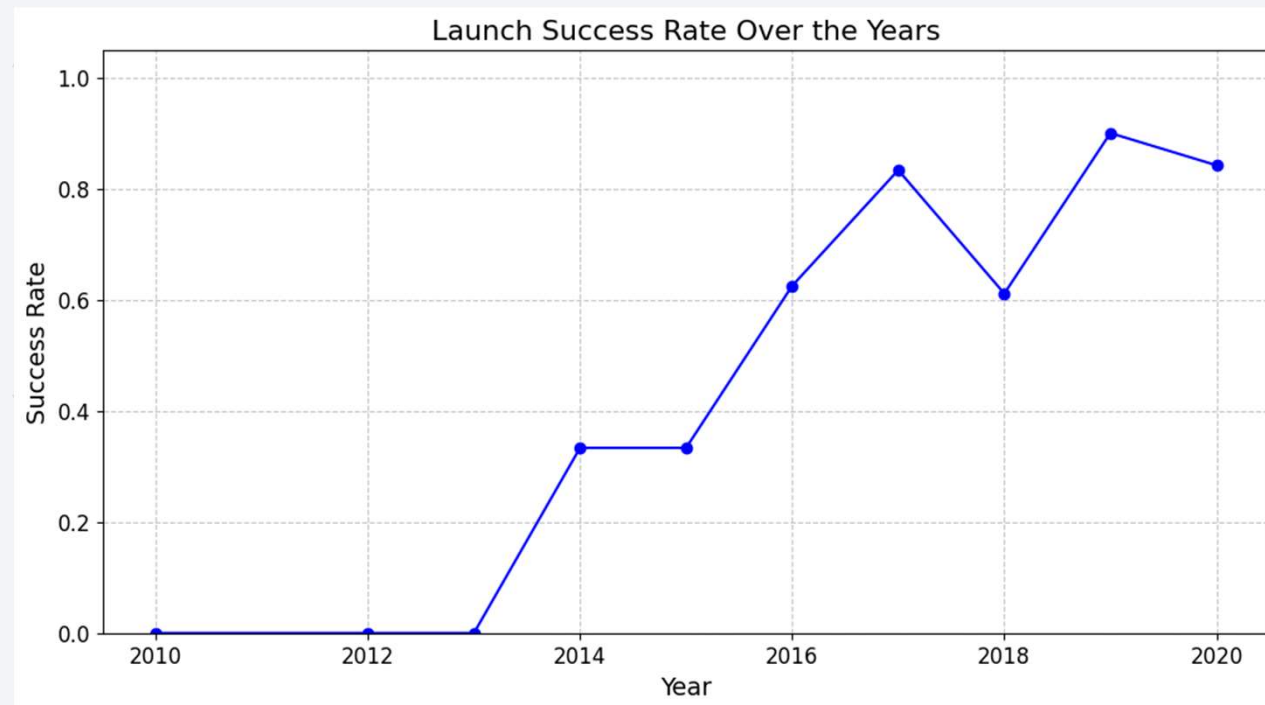
Payload vs. Orbit Type

- Only launches with orbit type ISS, PO and VLEO have more than 10000 kg payload mass



Launch Success Yearly Trend

- The average success rate improved over the years (except in 2018 and 2020)



All Launch Site Names

- The names of the unique launch sites

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- 5 records where launch sites begin with `CCA`

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

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Total Payload Mass

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- The total payload carried by boosters from NASA

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total_mass_carried_nasa

99980

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Average Payload Mass by F9 v1.1

- The average payload mass carried by booster version F9 v1.1

`total_mass_carried_f9v1p1`

38020

First Successful Ground Landing Date

- The dates of the first successful landing outcome on ground pad

Date
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- The names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- The total number of successful and failure mission outcomes

Total_Count
101

Boosters Carried Maximum Payload

- The names of the booster which have carried the maximum payload mass

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- The failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- The ranked count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

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Section 3

Launch Sites Proximities Analysis

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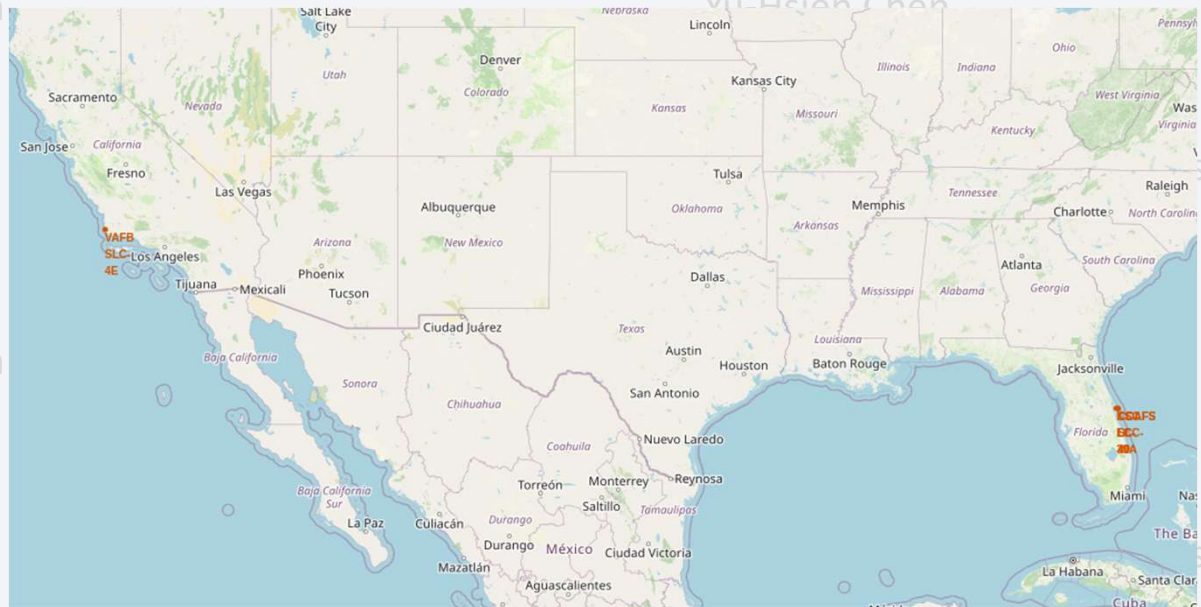
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Mark all launch sites on a map

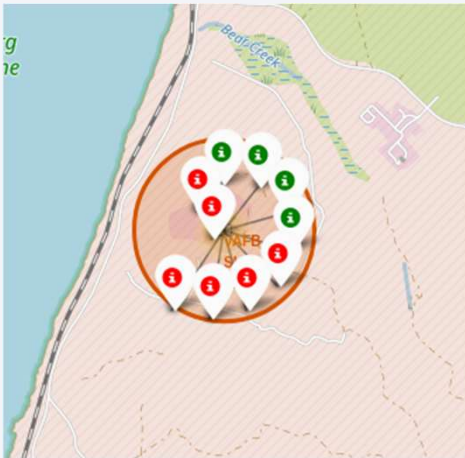
- The launch site VAFB SLC-4E is in the U.S. west (CA)
- The launch site CCAFS LC-40, CCAFS SLC-40 and KSC LC-39A are in the U.S. east (FL)



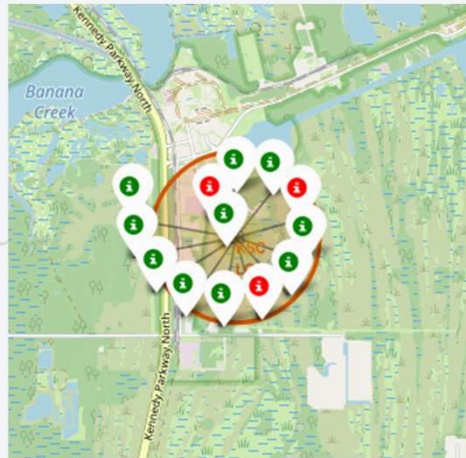
Mark the success/failed launches for each site on the map

- The greatest number of successful launches happened in site KSC LC-39A
- The greatest number of failed launches happened in site CCAFS LC-40

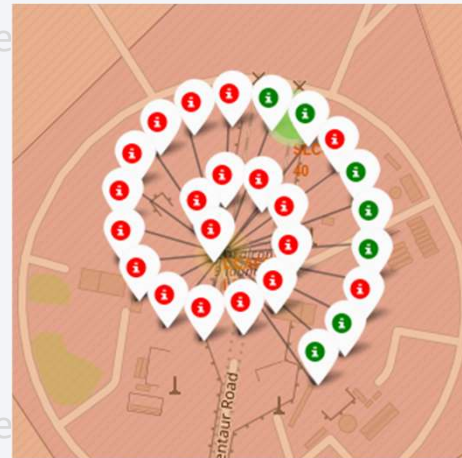
VAFB SLC-4E



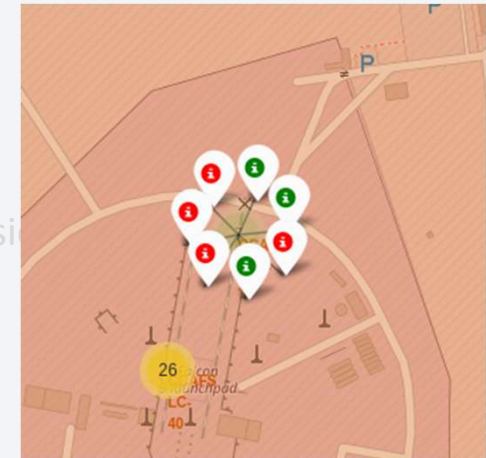
KSC LC-39A



CCAFS LC-40

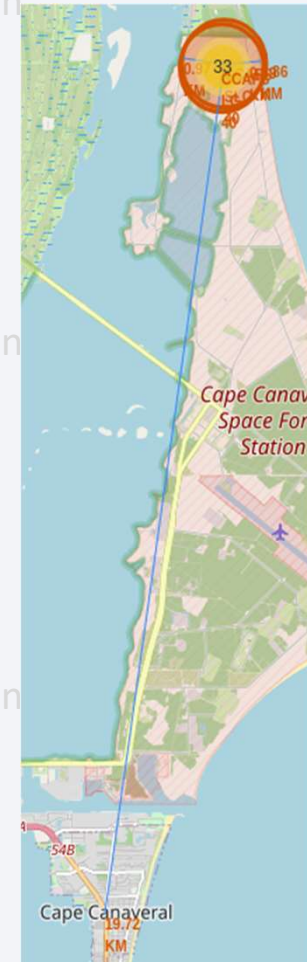


CCAFS SLC-40



Calculate the distances between a launch site to its proximities

- Distance between launch site CCAFS SLC-40 to:
 - Coastline – 0.86 km
 - Highway (Samuel C Phillips pkwy) - 0.59 km
 - Railway (Titan III road) - 0.97 km
 - City (Cape Canaveral) - 19.72 km
- The launch sites are closed to coastline, railway and highway. And the sites are far away from cities.



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Section 4

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Build a Dashboard with Plotly Dash

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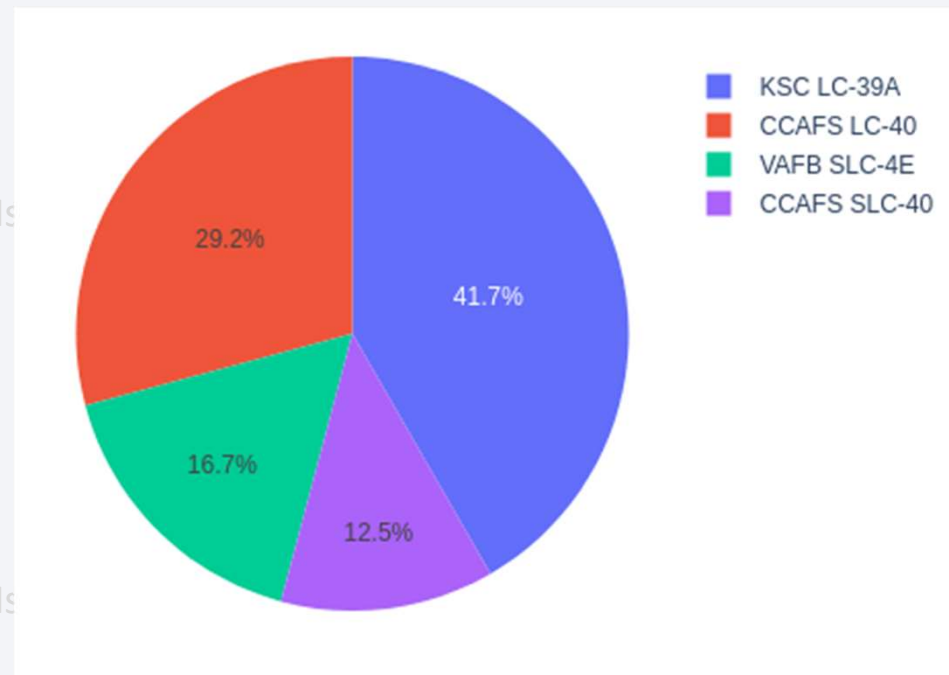
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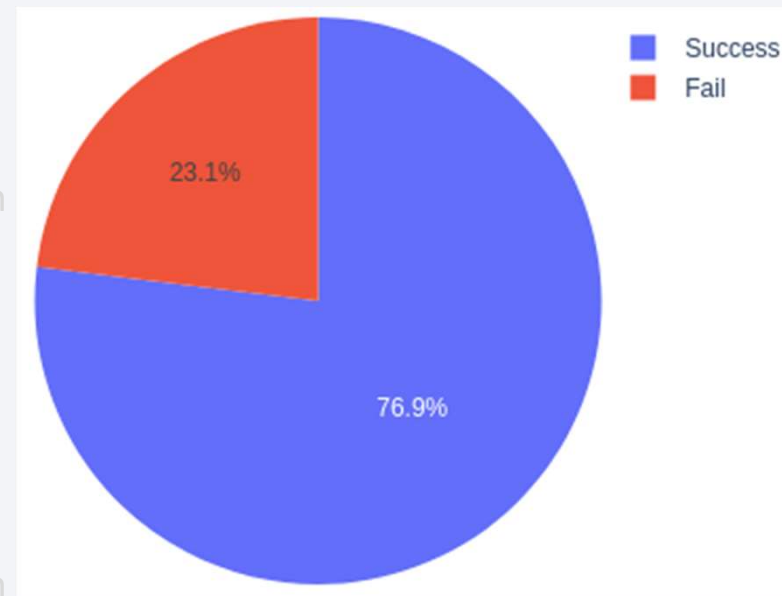
Success launches by launch sites

- The greatest number of successful launches happened in site KSC LC-39A
- The smallest number of successful launches happened in site CCAFS SLC-40



The launch site with highest launch success ratio

- The launch site with highest launch success ratio is KSC LC-39A (76.9 % success rate)



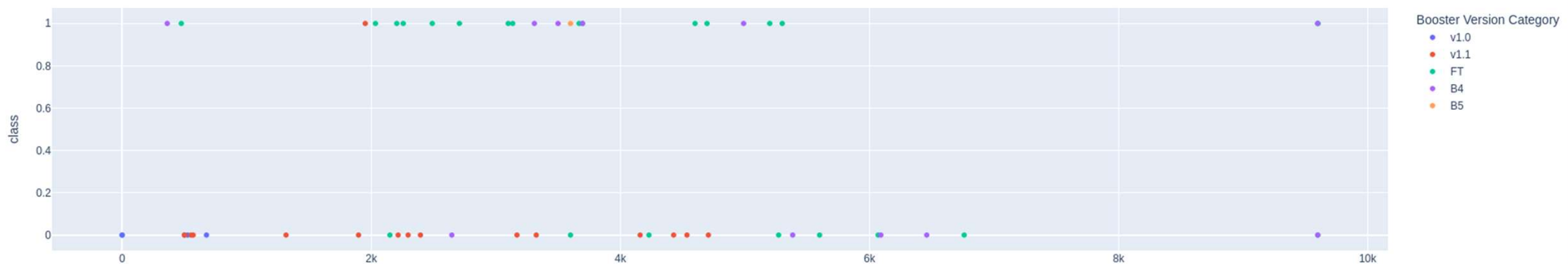
Payload vs. Launch Outcome scatter plot for all sites

- The booster version F9 FT have the largest success rate in payload range [2000, 6000]
- The booster version F9 v1.1 have the smallest success rate in payload range [0, 6000]

Payload range (Kg):



Correlation between Payload and Success for all sites



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Section 5

Predictive Analysis (Classification)

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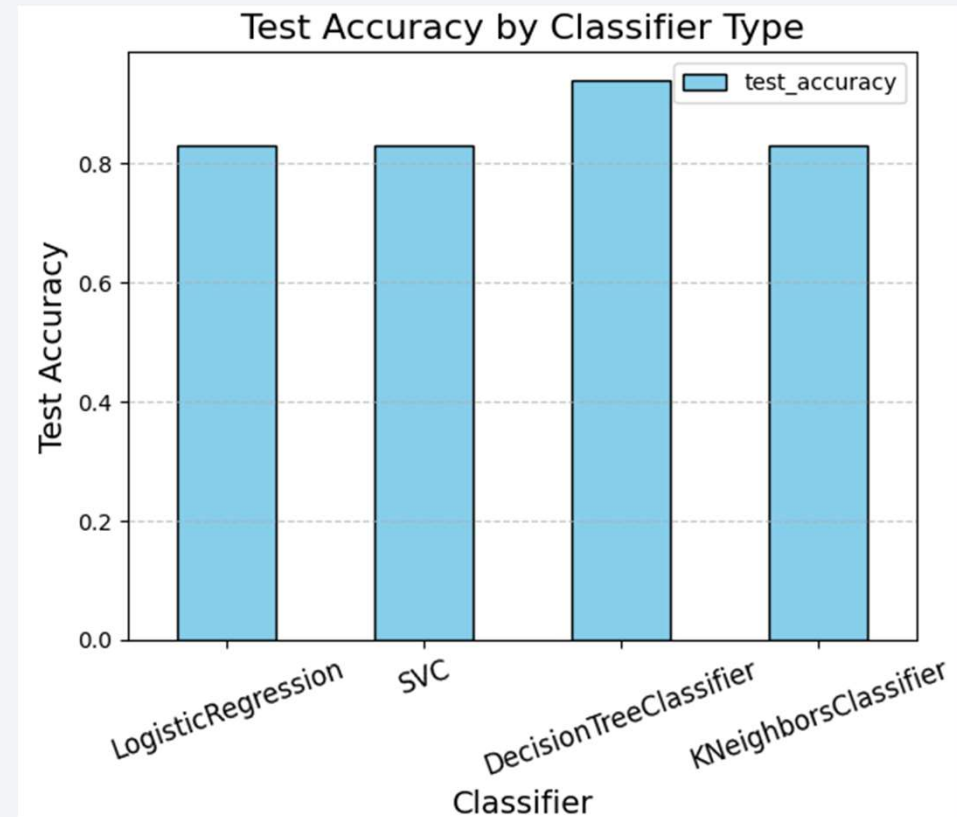
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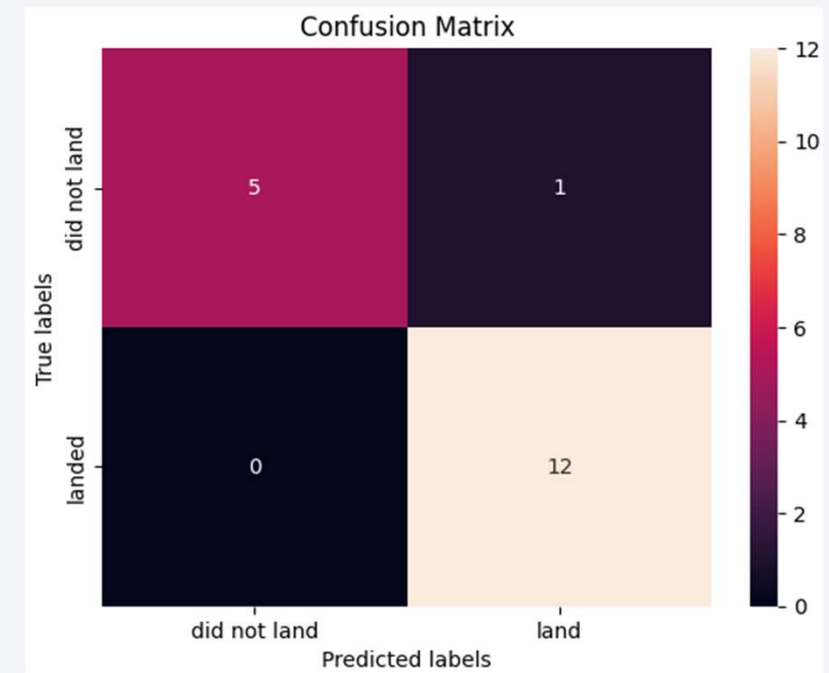
Classification Accuracy

- Decision tree classifier shows highest test accuracy (94 %)
- Logistic regression, SVC, K-nearest neighborhood classifiers show the lowest test accuracy (83 %)



Confusion Matrix

- Only 1 false positive cases (predicted landed but not landed).
- Accuracy = $(TP+TN)/(TP+TN+FP+FN) = (12+5)/(12+5+1) = 94 \%$



Conclusions

- It's possible to predict whether launch is successful using historical launch data with up to 94 % test accuracy.
- The decision tree classifier is most suitable in the 4 classifiers.

Appendix

- My codebase during this project: [AppliedDataScienceCapstone - github.com/Mimamsa](https://github.com/Mimamsa/AppliedDataScienceCapstone)

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Thank you!

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